



A method of quasi in-situ EBSD observation for microstructure and damage evolution in fatigue and dwell fatigue of Ti alloys

Chengqi Sun^{a,b,*}, Jian Sun^{a,b}, Weiqian Chi^c, Jiaxuan Wang^{a,b}, Wenjing Wang^{c,*}

^a State Key Laboratory of Nonlinear Mechanics, Institute of Mechanics, Chinese Academy of Sciences, Beijing 100190, China

^b School of Engineering Science, University of Chinese Academy of Sciences, Beijing 100049, China

^c Key Laboratory of Vehicle Advanced Manufacturing, Measuring and Control Technology (Beijing Jiaotong University), Ministry of Education, Beijing 100044, China

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ABSTRACT

A method of quasi in-situ electron backscattered diffraction observation is developed for investigating microstructure and damage evolution of Ti-6Al-4V ELI alloy under cyclic loadings. It indicates that the slip tends to occur in α grains with bigger Schmid factors (SFs) for basal slip or prismatic slip and twinning occurs in α grains with SFs not bigger than 0.2 for prismatic slip for both fatigue and dwell fatigue. The plastic strain caused by dwell stress could restrain the sliding or cleavage induced microcrack growth. Subgrain formation in α grains due to twinning and dislocation sliding are also captured during cyclic loadings.

1. Introduction

Titanium alloys have been widely used in ship and aviation fields (e.g., pressure hull of submersibles and fan discs of aeroengines) due to their high strength, low density, high corrosion resistance and high heat resistance [1,2]. Some of the components bear dwell fatigue loadings in service [3,4]. Therefore, a fully understanding of the dwell fatigue failure mechanism in Ti alloys is essential for the safe and reliable service of components. Similar to the normal fatigue, the dwell stress level [5,6], stress ratio [7,8] and microstructure of materials [7,9,10] could have great influence on the dwell fatigue behavior of Ti alloys. For relatively high dwell stress level, the maximum stress dwell leads to a significant decrease of fatigue life [7,11,12]. While for relatively low dwell stress level, the influence of maximum stress dwell on fatigue life can be negligible [4,7].

Some results have shown that the fracture surface exhibits facet feature in the crack initiation region for dwell fatigue of Ti alloys [3,13–15]. While the results for Ti6242 alloy with lamellar microstructure indicated that the crack initiation was attributed to the coalescence of cavities formed at α/β interfaces due to shearing, and the damage was mainly caused by pyramidal slip and enhanced by dwell loadings [16,17]. Regarding the facet formation, one viewpoint is that the quasi-cleavage facet in Ti alloys is due to the formation of slip within adjacent “hardly” oriented grains caused by dislocation pile-up in “softly” oriented grains at grain boundaries [15,18]. The essence of this

model is that the stress redistribution between the softly and hardly oriented α grains leads to the failure of hardly oriented grains at first. This result is supported by the calculation and simulation for considering the influence of different combinations of crystallographic orientation relationship in adjacent α grains on the local stress and stress redistribution between grains [19–21].

Recently, the study for the dwell fatigue of Ti-6Al-4V ELI alloy showed that the fracture surface of specimens presented three different failure modes: fatigue failure, mixed failure and ductile failure [8]. The fatigue failure mode occurs when the fatigue crack initiation and growth dominate, and the ductile failure appears when the plastic deformation dominates. This indicates that the dwell fatigue mechanism is strongly related to the failure mode in Ti alloys, and the investigation of dwell fatigue mechanism should be combined with the failure mode. However, the existing studies are whether mainly on the mechanism of fatigue failure or not referring to the failure mode under dwell fatigue loadings, and there are few results available for considering the competition of different failure mechanism in microstructural scale.

In this paper, a method of quasi in-situ electron backscattered diffraction (EBSD) observation is at first proposed by the combination of conventional MTS machine and EBSD observation. Then, it is used to investigate the microstructure and damage evolution of Ti-6Al-4V ELI alloy used for pressure hulls of submersibles under fatigue and dwell fatigue loadings. The new method successfully achieves the characterization on crystallographic orientation and microstructure characteristic

* Corresponding authors.

E-mail addresses: scq@lnm.imech.ac.cn (C. Sun), wjwang@bjtu.edu.cn (W. Wang).

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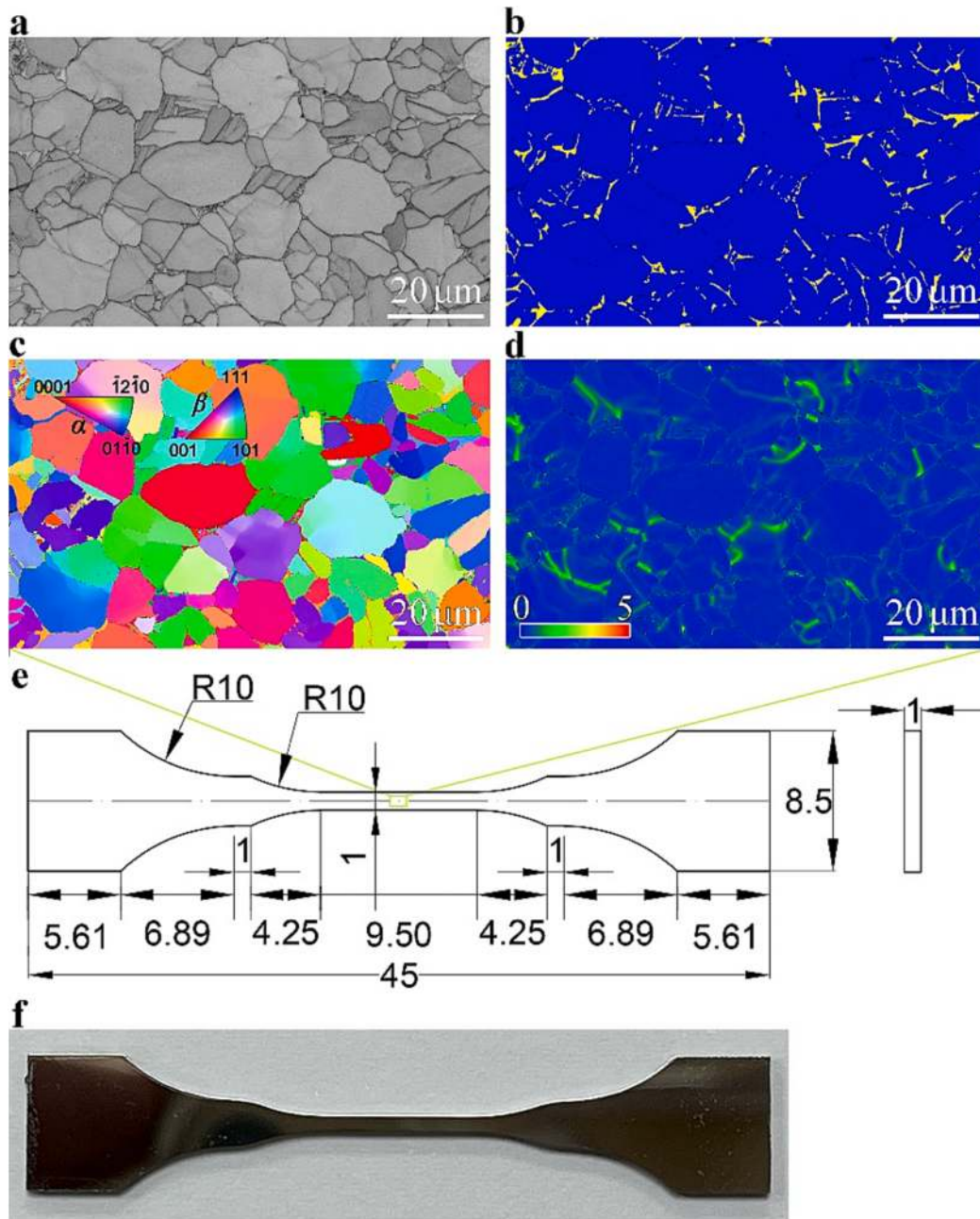


Fig. 1. EBSD results of microstructure. a-d: Band contrast (BC) map, phase map (α phase: blue, β phase: yellow), inverse pole figure (IPF), and KAM map, respectively. e: Shape and geometry of specimens for fatigue and dwell fatigue tests (mm). f: Picture of actual specimen. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in the labelled regions before and after a certain amount of fatigue or dwell fatigue cycles. By the comparison of microstructure and damage under fatigue and dwell fatigue loadings, the influence of maximum stress dwell and dwell time on fatigue behavior is further analyzed and discussed. It is found that a certain degree of maximum stress dwell favors the activation of slip system, but it is not the case that the greater the deformation the more favorable the slip system activation. The maximum stress dwell could favor the crack nucleation due to sliding or cleavage, and it could also restrain the growth of microcrack induced by sliding or cleavage, which is related to the dwell stress level and dwell time. The dwell fatigue mechanism of Ti alloys is also discussed in details.

2. Materials and methods

2.1. Materials

The material was cut from a forged and then rolled plate made of Ti-6Al-4V ELI alloy [8], and the specimen axis is along the rolling direction. The microstructure is shown in Fig. 1a-1c, which approximately consists of equiaxed α phase and lamellar β_{tran} . The associated kernel average misorientation (KAM) map is shown in Fig. 1d, which indicates the locally nonuniform dislocation density [22,23]. The tensile and yield stresses are 879 MPa and 831 MPa [8], respectively.

Table 1

Testing information of specimens.

Loading type	Maximum stress/MPa	Dwell time $d_{t,max}$ /s	Minimum stress/MPa	Fatigue life/cyc	Failure mode
Dwell fatigue	815	10	0	883	Ductile failure
Dwell fatigue	815	120	0	199	Ductile failure
Fatigue	815	0	0	10,686	Fatigue failure

2.2. Experimental methods

The fatigue tests were all conducted on the MTS machine at room temperature in air. The loading type was axial loading with loading control mode. The trapezoidal wave was used for dwell fatigue specimens and the triangular wave was used for fatigue specimens. For dwell fatigue tests, the dwell time of 10 s and 120 s of the maximum stress was used to investigate the effect of dwell time on the fatigue behavior. The rise and fall time were 2 s for both dwell fatigue and fatigue tests. The dwell fatigue specimen with dwell time of 10 s was at first tested, and then the fatigue specimen was tested with an intermediate stop at the fatigue life of the dwell fatigue specimen with dwell time of 10 s, which was used to study the influence of maximum stress dwell and fatigue loading cycle on the microstructure and damage evolution. Finally, the dwell fatigue specimen with dwell time of 120 s was tested, and it was stopped at several different dwell fatigue cycles, which was used to

study the influence of dwell fatigue loading cycle on the microstructure and damage evolution. To observe the microstructure evolution at specimen surface, the plate specimen was designed and used for both the fatigue and dwell fatigue tests. The shape and geometry of the specimen and its actual picture are shown in Fig. 1e and 1f, respectively. The observed surface and its opposite were ground and polished using abrasive papers (180 grits, 500 grits, 1000 grits, 3000 grits, and 5000 grits), and the observed surface was further polished with the non-crystallizing Colloidal Silica polishing suspension with a particle size of 0.02 μm .

To capture the microstructure evolution, a quasi in-situ EBSD observation is proposed by the combination of the MTS machine and EBSD observation. The method is described as follows: i) Some regions are labelled before (dwell) fatigue tests or after a number of (dwell) fatigue cycles by indentation on the specimen surface; ii) The specimen with labelled regions is tested to the given cycles, and then removed from the machine; iii) The specimen surface is washed with clean water, and then the detergent is used to wash the gauge section of the specimen with moderate force in one direction for about 10 times, and then the gauge section is washed with moderate force in the same direction for about 10 times under clean water, and finally the compressive air is used to blow dry the specimen surface immediately after cleaning; iv) The specimen surface especially the labelled regions are observed by EBSD. In this way, the microstructure and damage evolution after a number of loading cycles could be captured by the EBSD analysis and the associated scanning electron microscope (SEM) observation.

The fracture surfaces were observed by SEM for identifying the

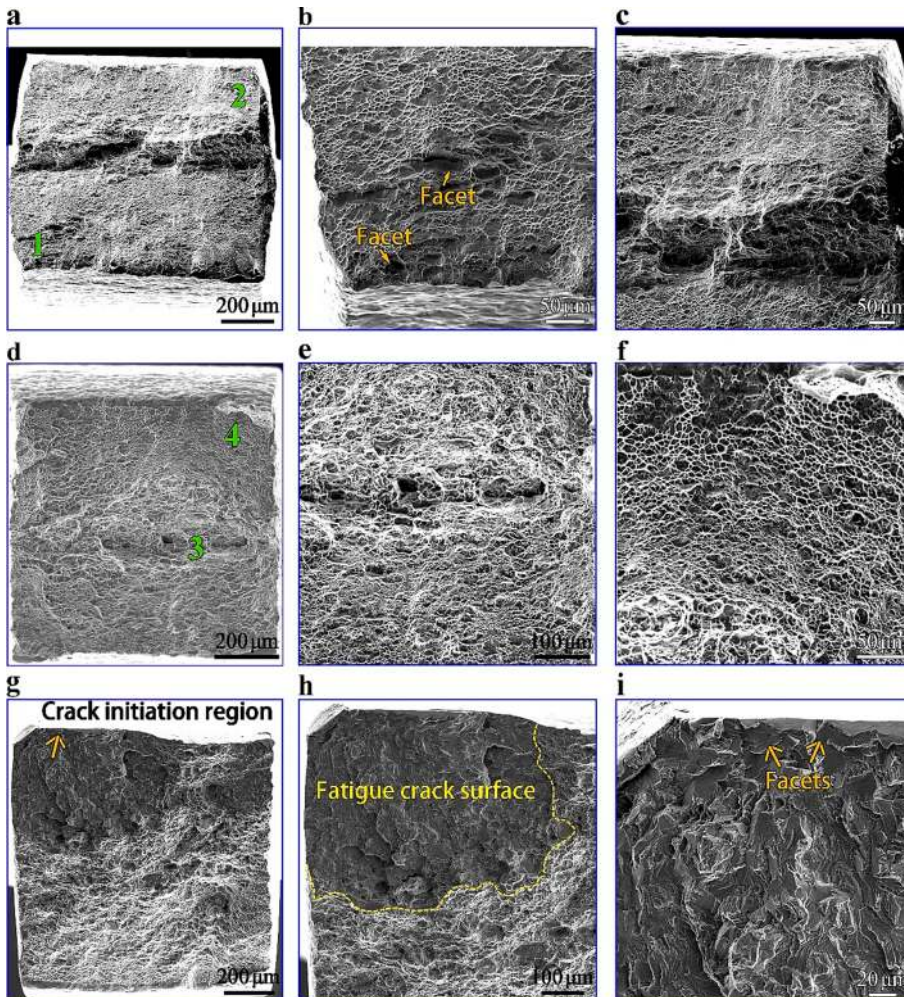


Fig. 2. Fracture surface morphologies of specimens. a-c: Dwell fatigue specimen with $d_{t,max} = 10$ s, $\sigma_{max} = 815$ MPa, and $N_f = 883$ cycles; b and c are close-ups of regions 1 and 2 in a, respectively. d-f: Dwell fatigue specimen with $d_{t,max} = 120$ s, $\sigma_{max} = 815$ MPa, and $N_f = 199$ cycles; e and f are close-ups of regions 3 and 4 in d, respectively. g-i: Fatigue specimen, $\sigma_{max} = 815$ MPa and $N_f = 10686$ cycles; h and i are close-ups of the fatigue crack initiation and growth region in g and h, respectively.

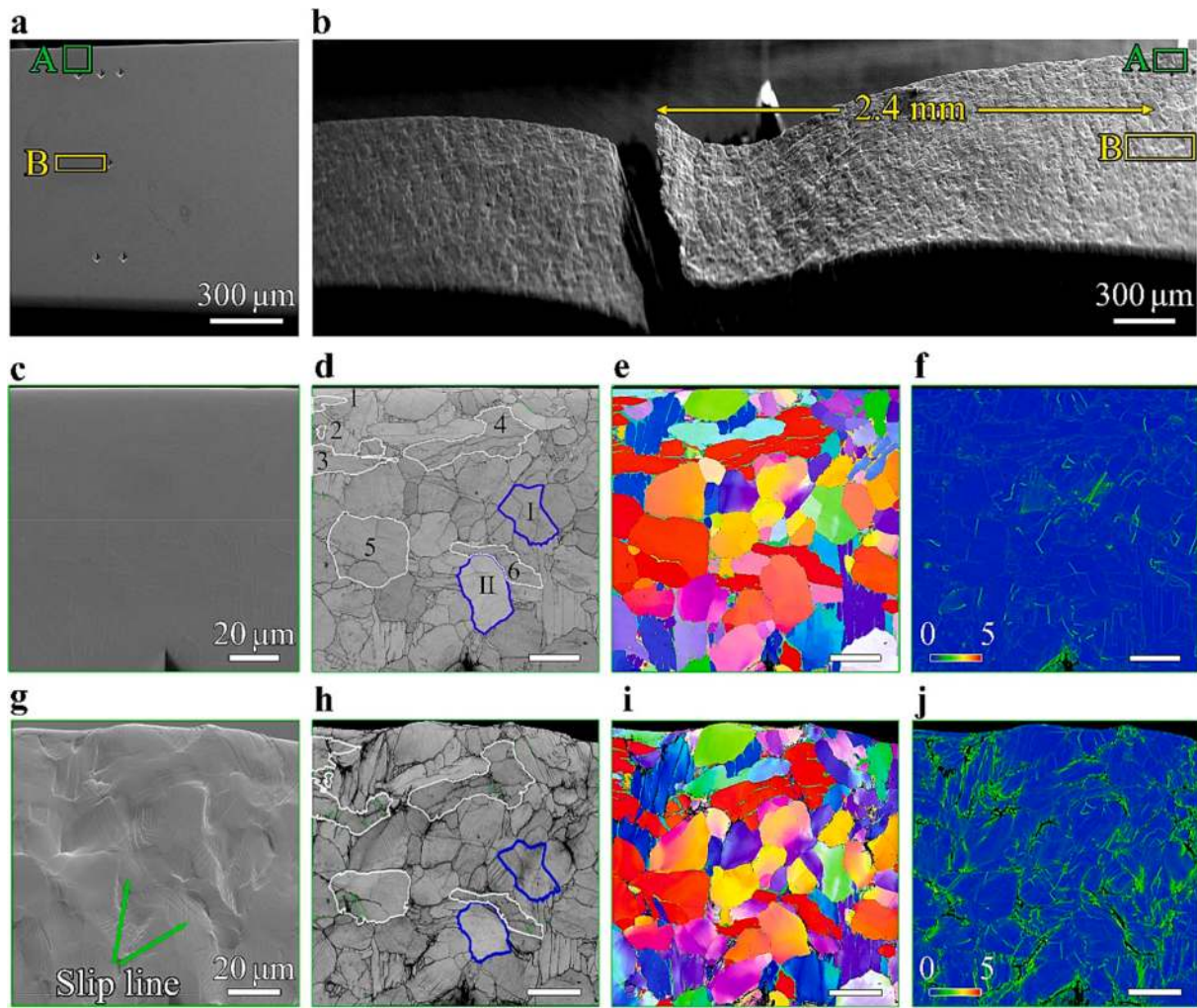


Fig. 3. Microstructure evolution during dwell fatigue loadings ($\sigma_{\max} = 815$ MPa, $d_{t,\max} = 10$ s and $N_f = 883$ cycles). a and b: SEM observation of part of the initial specimen and the same specimen after 883 dwell fatigue cycles. c-f and g-j: Enlarged SEM image of the labelled region A and the associated BC map, IPF and KAM map of the initial specimen and the specimen after 883 dwell fatigue cycles, respectively. The α grain enclosed by the white line in d and h is with twinning occurrence, and the green lines denote the $\{10\bar{1}2\}$ twin boundary. The α grain enclosed by the blue line in d and h is with bigger intracrystal deformation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

fracture surface morphologies and failure modes.

3. Results and analyses

3.1. Fatigue performance and fracture surface morphology

The dwell fatigue specimen with dwell time of 10 s was tested continuously, and it failed at 883 cycles. The dwell fatigue specimen with dwell time of 120 s was with intermediate stops at 74 cycles, 126 cycles and 178 cycles, and then it was tested continuously until failure. The fatigue specimen was with an intermediate stop at 883 cycles, and then it was tested continuously until failure. The fatigue life and failure mode of the tested specimens are shown in Table 1. It is seen from Table 1 that the dwell of the maximum stress significantly reduces the fatigue life, and the dwell fatigue life decreases with the increase of dwell time.

The dwell fatigue specimens with dwell time of 10 s and 120 s both exhibit the ductile failure mode, similar to that observed for the dwell fatigue of the column specimen of the same titanium alloy [8]. In this failure mode, there is no or very small fatigue crack region on the fracture surface (Fig. 2a-2f), and the failure is dominated by the plastic deformation. While the fatigue specimen presents a big fatigue crack

initiation and crack growth region on the fracture surface (Fig. 2g-2i). This indicates that the fatigue crack initiation and growth is the main factor for the failure of the specimen, i.e., the fatigue specimen exhibits fatigue failure mode. The facet feature is observed in the crack initiation region of the fatigue specimen (Fig. 2i), and it is also observed at the local fracture surface of the dwell fatigue specimen with dwell time of 10 s (Fig. 2b).

3.2. SEM and EBSD observation of specimen surface after dwell fatigue loadings

3.2.1. Dwell fatigue with dwell time of 10 s

Figs. 3 and 4 show the SEM and EBSD results of two labelled regions 2.4 mm away from the fracture surface for the dwell fatigue specimen with dwell time of 10 s before and after dwell fatigue test. It is observed from the SEM images in Fig. 3g and Fig. 4b that a lot of slip traces appear on the specimen surface, and the specimen surface exhibits obvious concave and convex feature. EBSD analysis indicates that the deformation twinning ($\{10\bar{1}2\} \langle 10\bar{1}1 \rangle$ (extension twin) appears in α grains after dwell fatigue loadings, as shown in Fig. 3h and Fig. 4d. The twins in this paper are identified through the analysis of grain boundaries with a tolerance of $\pm 5^\circ$ deviation via HKL CHANNEL5 software [24–27].

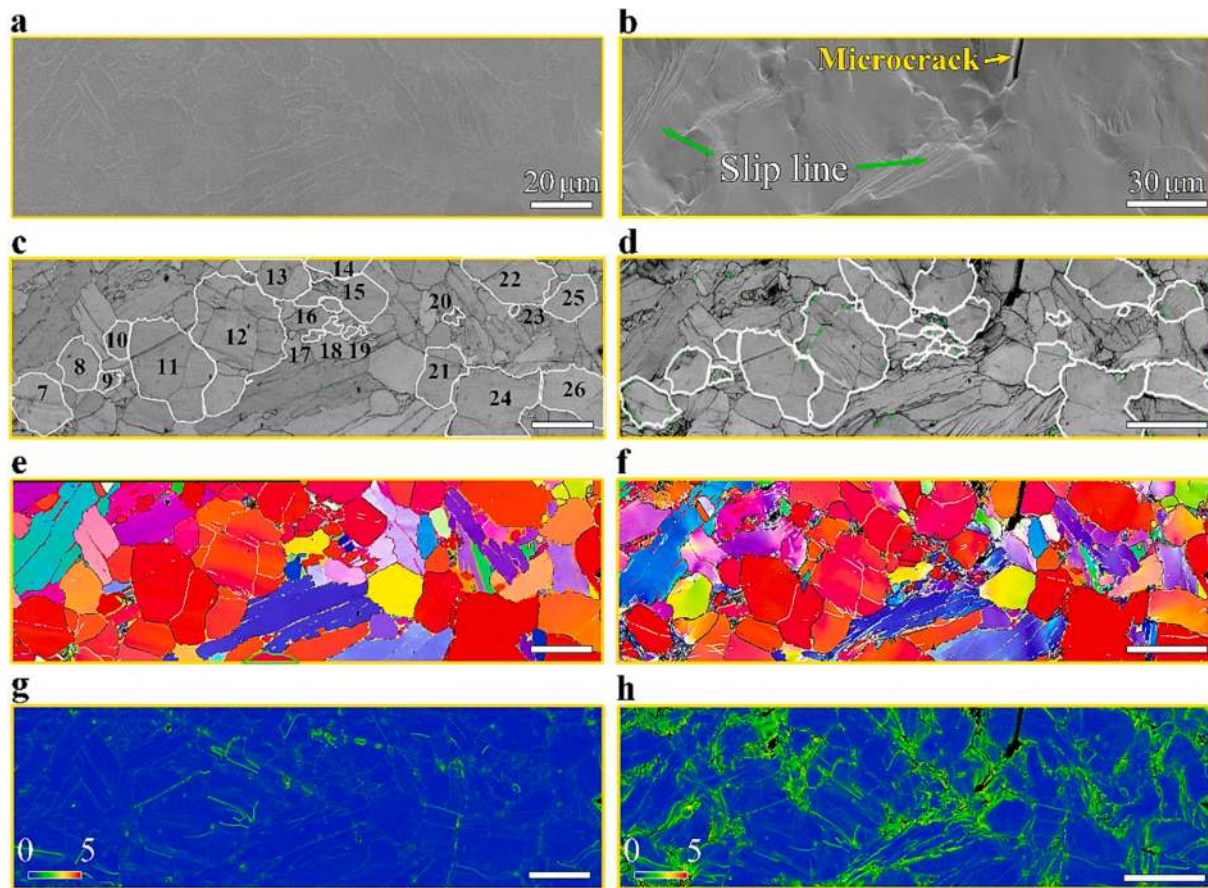


Fig. 4. SEM and EBSD results of the labelled region B in Fig. 3a. a, c, e and g: Enlarged SEM image of the labelled region B and the associated BC map, IPF and KAM of the initial specimen, respectively. b, d, f and h: Enlarged SEM image of the labelled region B and the associated BC map, IPF and KAM after 883 dwell fatigue cycles, respectively. The α grain enclosed by the white line in c and d is with twinning occurrence, and the green lines denote the $\{10\bar{1}2\}$ twin boundary. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Further, a microcrack is observed in the labelled region B (Fig. 4b). The microcracks are also found in the other regions very near the fracture surface (Fig. 5b), a little far from the fracture surface (Fig. 5c and 5d) and far from the fracture surface (Fig. 5e and 5g) after dwell fatigue loadings with dwell time of 10 s. Some cavities are also observed a little far from the fracture surface, as shown in Fig. 5f.

3.2.2. Dwell fatigue with dwell time of 120 s

Figs. 6 and 7 shows the SEM and EBSD results of two labelled regions after 74, 126, 178 and 199 dwell fatigue cycles. The labelled region A is about 360 μm away from the fracture surface, and the labelled region B is about 2.1 mm away from the fracture surface, as indicated in Fig. 6a. It is seen from Fig. 6b1-6d1 and Fig. 6b4-6d4 that the deformation of region A near to the final fracture surface seems to be no change after 74, 126 and 178 cycles from both SEM images and KAM maps. It also seems to be no change from SEM images from 178 to 199 cycles, but there are some changes from KAM maps, as shown in Fig. 6d1, Fig. 6d4, Fig. 6e1 and Fig. 6e4. While for the region B far away from the fracture surface, it seems to be no change from both SEM images and KAM maps from 74 to 199 cycles, as shown in Fig. 7a1-7a4 and Fig. 7d1-7d4. This should be attribute to that the deformation near the final fracture region increases significantly with increasing the loading cycles when the specimen begins to fracture. The results in Figs. 6 and 7 indicate that the deformation of the specimen generally do not change significantly after a certain amount of fatigue cycles. But with the further increase of dwell fatigue cycles, the deformation becomes to be much bigger in some local regions. When it reaches a value, these local regions fail and induce the final failure. This process is inferred by the KAM values of the two

labelled regions near to and far from the fracture surface after different dwell fatigue cycles in Figs. 6 and 7. It is noted that the two labelled regions in Fig. 6a are chosen after 74 dwell fatigue cycles, which have relative bigger concave convex features in SEM images.

The deformation twinning is also observed for the dwell fatigue with dwell time of 120 s, and they are all extension twins, as shown in Fig. 6b2-6e2 and Fig. 7a2-7d2. A comparison of the observed regions A and B after different dwell fatigue cycles indicates that some twinning has appeared at 74 cycles and some new twinning (e.g., the twins in grains 5 and 7 in Fig. 6b2 and 6c2) occurs with the increase of dwell fatigue cycles. This indicates that whether the occurrence of twinning depends on both the microstructure of material and the loading condition. Moreover, it is observed that some twins grow and the others do not change with the increase of dwell fatigue cycles.

By using SEM, most of the specimen surfaces were examined after the fracture of the specimen under dwell fatigue loadings with dwell time of 120 s. The results of some regions are shown in Fig. 8. For the regions a little far or far from the fracture surface, the ones with relative bigger concave convex features in SEM images are chosen. Fig. 8 indicates that the deformation is generally small in the regions far from the fracture surface (Fig. 8b and 8c) compared with the regions near or a little far from the fracture surface (Fig. 8d-8i). Some microcracks or cavities are observed near or a little far from the fracture surface, as shown in Fig. 8d-8i. However, there is no microcrack or cavity observed on the specimen surface far from the fracture surface. This indicates that the microcrack or cavity near the fracture surface should be mainly due to the increase of the local high strain when the specimen is going to fail.

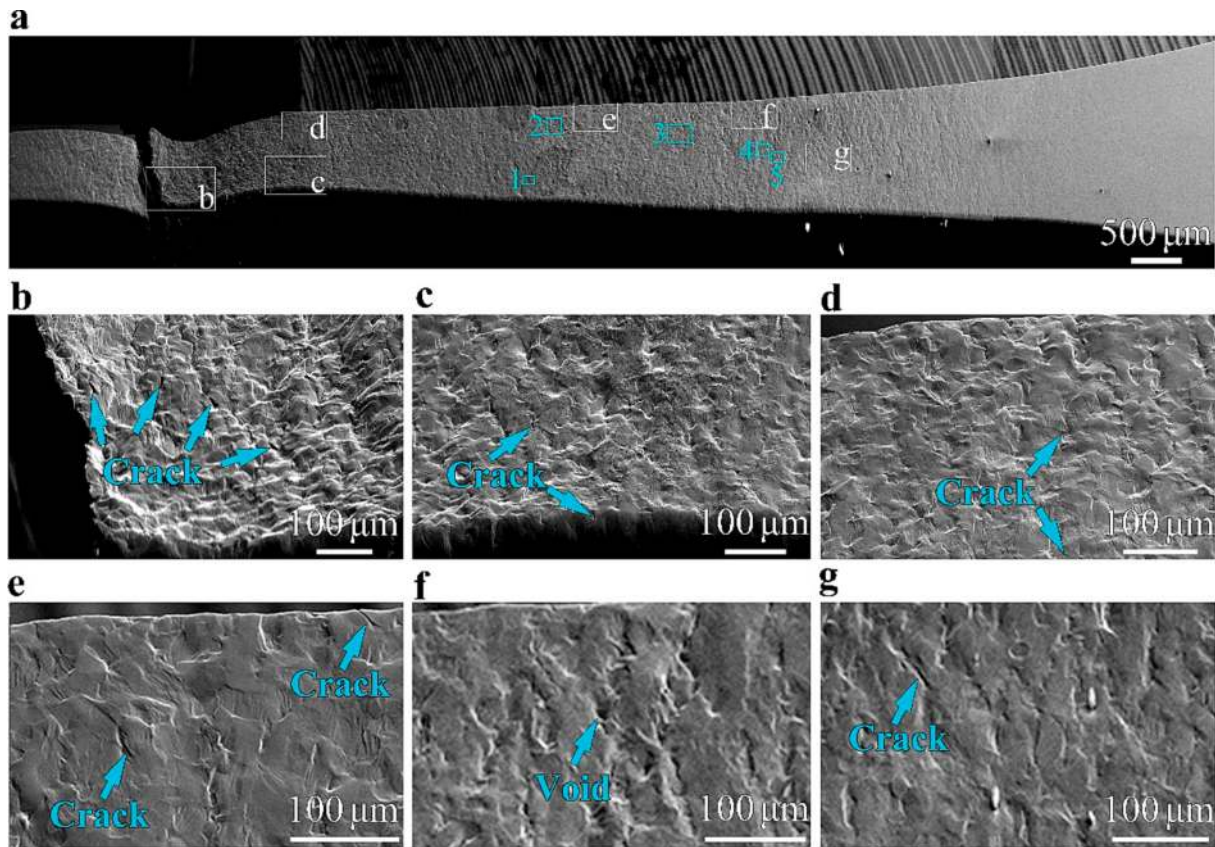


Fig. 5. SEM observation of specimen surface after 883 dwell fatigue cycles ($\sigma_{\max} = 815$ MPa, $d_{t,\max} = 10$ s and $N_f = 883$ cycles). a: SEM observation of part of the specimen surface. b-g: Close-ups of the regions b-g in a, respectively. The regions 1 to 5 in a denote where the microstructure and crack is further analyzed in Sec. 4.5.

3.3. SEM and EBSD observation of specimen surface after fatigue loadings

Figs. 9 and 10 show SEM and EBSD results of two labelled regions of the specimen during fatigue loadings. The two labelled regions are about 3.3 mm far away from the fracture surface. Twice quasi in-situ observations are conducted. One is after 883 cycles, and the other is after 10,686 cycles (i.e., the failure of the specimen). SEM results in Fig. 9b1-9d1 and Fig. 10a-10c indicate that the slip traces emerge in some α grains after 883 cycles and some of these slip traces deepen after 10,686 cycles. The twinning is found in a few α grains after 883 cycles and the number of α grains with twinning occurrence increases from 883 to 10,686 cycles, as shown in Fig. 9b2-9d2 and Fig. 10g-10i. It is also observed that the number of twins occurred in α grains increases with increasing the loading cycle.

After 10,686 fatigue loading cycles (i.e., the failure of the specimen), the microstructure characteristics are examined by SEM for most of the specimen surface, as shown in Fig. 11a. Many microcracks and bigger cracks exceeding 100 μm are observed, as shown in Fig. 11b-11g. A comparison with the failed specimen under dwell fatigue loadings in Figs. 5 and 8 indicates that the maximum stress dwell has significant influence on fatigue crack nucleation and growth. For the maximum stress dwell of 120 s, no crack is observed on the specimen surface far from the fracture surface after the specimen fails. While for the maximum stress dwell of 10 s, only some microcracks are observed on the specimen surface, and these microcracks are in one or two grain size.

4. Discussions

4.1. Deformation under fatigue and dwell fatigue loadings

The KAM values in Figs. 3 and 4 for the fatigue specimen and the

ones in Figs. 9 and 10 for the dwell fatigue specimen after 883 cycles indicate that the dwell of maximum stress increases the cumulative plastic deformation under the same loading cycles, which shows the evidence of the microstructural scale deformation for the increase of the macroscopic deformation due to the dwell of maximum stress in previous studies [8]. The maximum stress dwell increases the geometrically necessary dislocations in local regions and results in bigger deformation in grains or grain boundaries in comparison with the fatigue loading.

For both fatigue and dwell fatigue loadings, the increment of misorientation and the subgrain formation are captured in α grains by the quasi in-situ observation, as shown in Fig. 12. It is seen that the misorientation in α grains could increase significantly after a number of loading cycles, which is irrespective of fatigue or dwell fatigue loadings. This indicates that the cumulative plastic strain during the fatigue or dwell fatigue loading could lead to the formation of subgrains in α grains with preferable orientation. In particular, the formation of high angle grain boundaries (larger than 10°) in the α grain is observed for the sample during the dwell fatigue loading with dwell time of 10 s (Fig. 12d2), which provides with the evidence of the smaller grain formation due to the dislocation sliding and dislocation interaction during cyclic loadings.

4.2. Slip under fatigue and dwell fatigue loadings

Slip is an important characteristic of deformation. For a further understanding of the occurrence of slip system, Fig. 13a-13c shows the activation of the identifiable single slip systems in α grains at specimen surface after the fracture of the specimen under fatigue and dwell fatigue loadings, and Fig. 13d-13f shows the associated Schmid factors (SFs) for basal slip and prismatic slip of α grains in Fig. 13a-13c, respectively. The slip systems are at first predicted by MTEX in MATLAB software [28,29] and then confirmed by the slip traces on SEM images. For the fatigue and

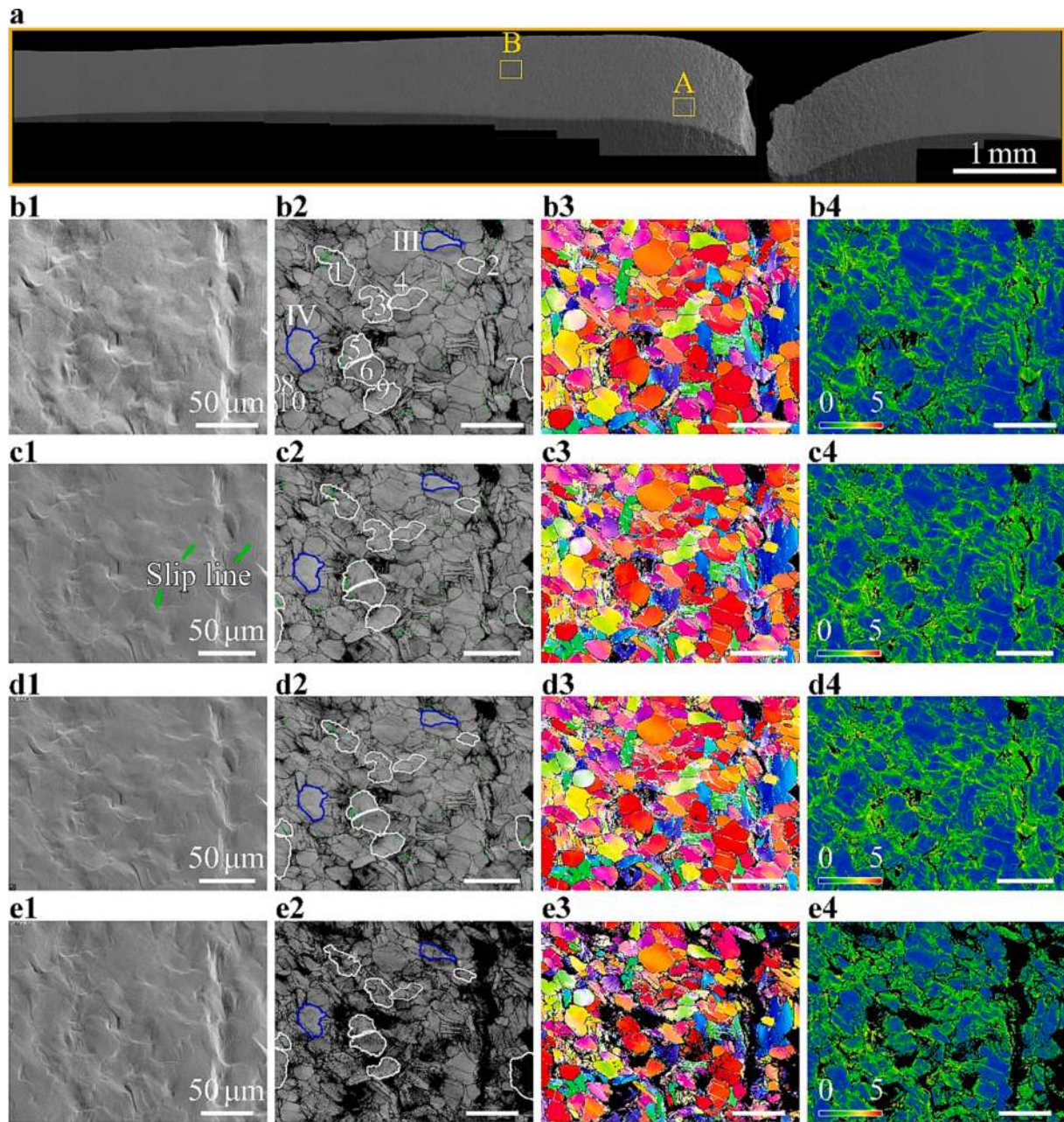


Fig. 6. Microstructure evolution during dwell fatigue loadings ($\sigma_{\max} = 815$ MPa, $d_{t,\max} = 120$ s and $N_f = 199$ cycles). a: SEM observation of part of the specimen after failure. b1-b4, c1-c4, d1-d4 and e1-e4: Enlarged SEM image of the labelled region A and the associated BC map, IPF and KAM map after 74, 126, 178 and 199 dwell fatigue cycles, respectively. The α grain enclosed by the white line in b2-e2 is with twinning occurrence, and the green lines denote the $\{10\bar{1}2\}$ twin boundary. The α grain enclosed by the blue line in b2-e2 is with bigger intracrystal deformation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

dwell fatigue with dwell time of 10 s, the SFs for basal slip and prismatic slip are calculated from the crystallographic orientation of the original α grains due to the EBSD observation of these labelled regions before fatigue and dwell fatigue tests. For the dwell fatigue with dwell time of 120 s, they are calculated from the crystallographic orientation of the α grains after 74 loading cycles.

It is seen from Fig. 13a and 13d that the basal slip system dominates for the activation of single slip systems in α grains under fatigue loadings. The SF for basal slip is bigger than the one for prismatic slip for the α grains with the activation of basal slip and most of the SFs for basal slip are larger than 0.35. The prismatic slip and pyramidal slip ($Py_2 < c + a >$) activate in a few α grains. The SF for prismatic slip is bigger than the SF for basal slip except for one grain for the α grains with the activation

of prismatic slip and the SFs for prismatic slip are all bigger than 0.35. The pyramidal slip activates in α grains with bigger SF for pyramidal slip. For the dwell fatigue with dwell time of 10 s, both basal slip and prismatic slip systems dominate, as shown in Fig. 13b and 13e. The SF for basal slip is bigger than the one for prismatic slip except for two grains for the α grains with the activation of basal slip and most of the SFs for basal slip are larger than 0.35. The SF for prismatic slip is bigger than the SF for basal slip except for three grains for the α grains with the activation of prismatic slip and the SFs for prismatic slip are often bigger than 0.35. The pyramidal slip activates in a few α grains, and most of the SFs are bigger for pyramidal slip. Similar to the dwell fatigue with dwell time of 10 s, the basal slip and prismatic slip also dominate for the dwell fatigue with dwell time of 120 s, as shown in Fig. 13c and 13f. The SF for

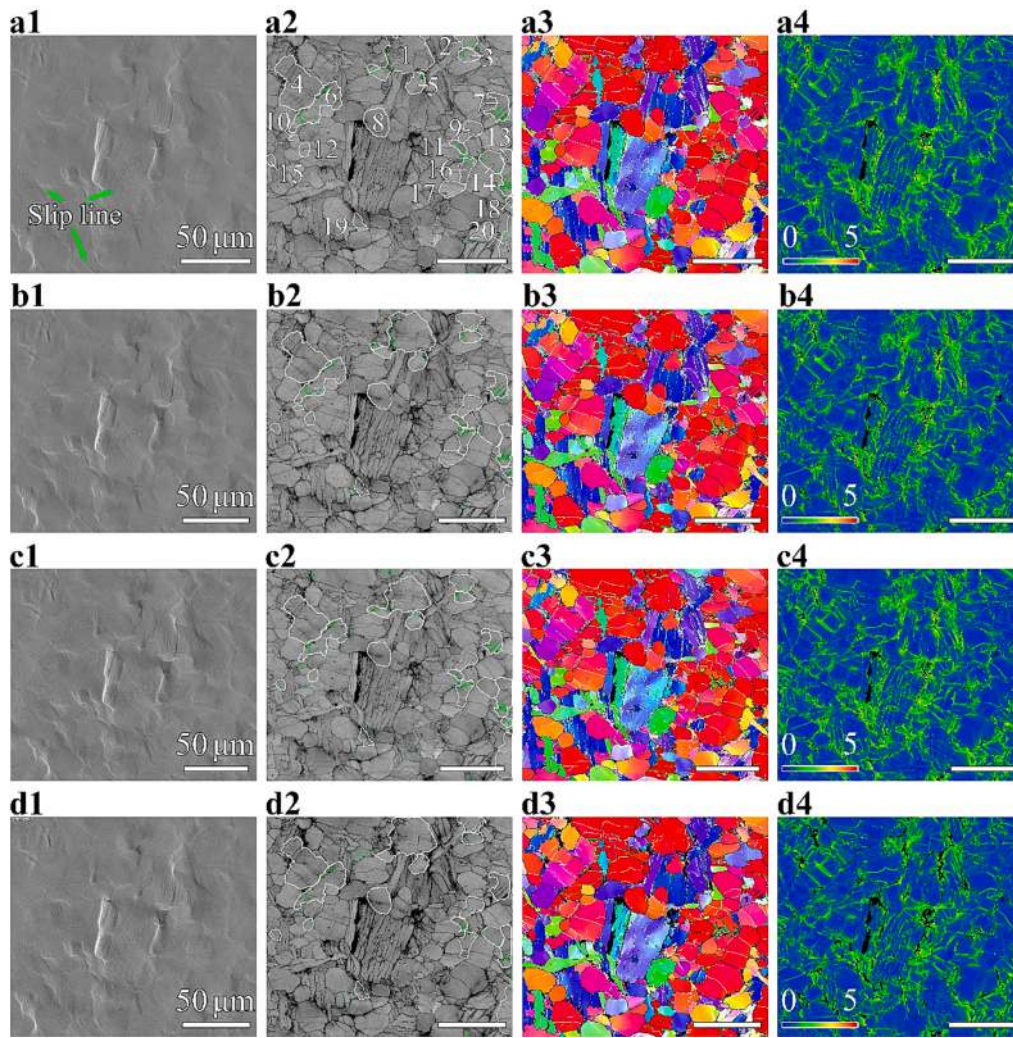


Fig. 7. SEM and EBSD results of the labelled region B in Fig. 6a during dwell fatigue loadings ($\sigma_{\max} = 815$ MPa, $d_{t,\max} = 120$ s and $N_f = 199$ cycles). a1-a4, b1-b4, c1-c4 and d1-d4: Enlarged SEM image and the associated BC map, IPF and KAM map after 74, 126, 178 and 199 dwell fatigue cycles, respectively. The α grain enclosed by the white line in a2-d2 is with twinning occurrence, and the green lines denote the $\{10\bar{T}2\}$ twin boundary. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

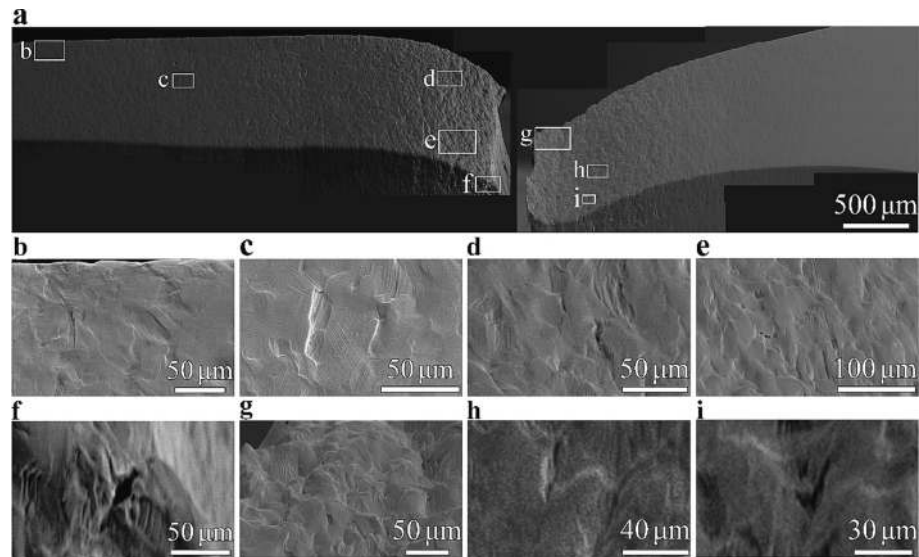


Fig. 8. SEM observation of specimen surface after 199 dwell fatigue cycles ($\sigma_{\max} = 815$ MPa, $d_{t,\max} = 120$ s and $N_f = 199$ cycles). a: Part of the specimen surface. b-i: Close-ups of the regions b-i in a, respectively.

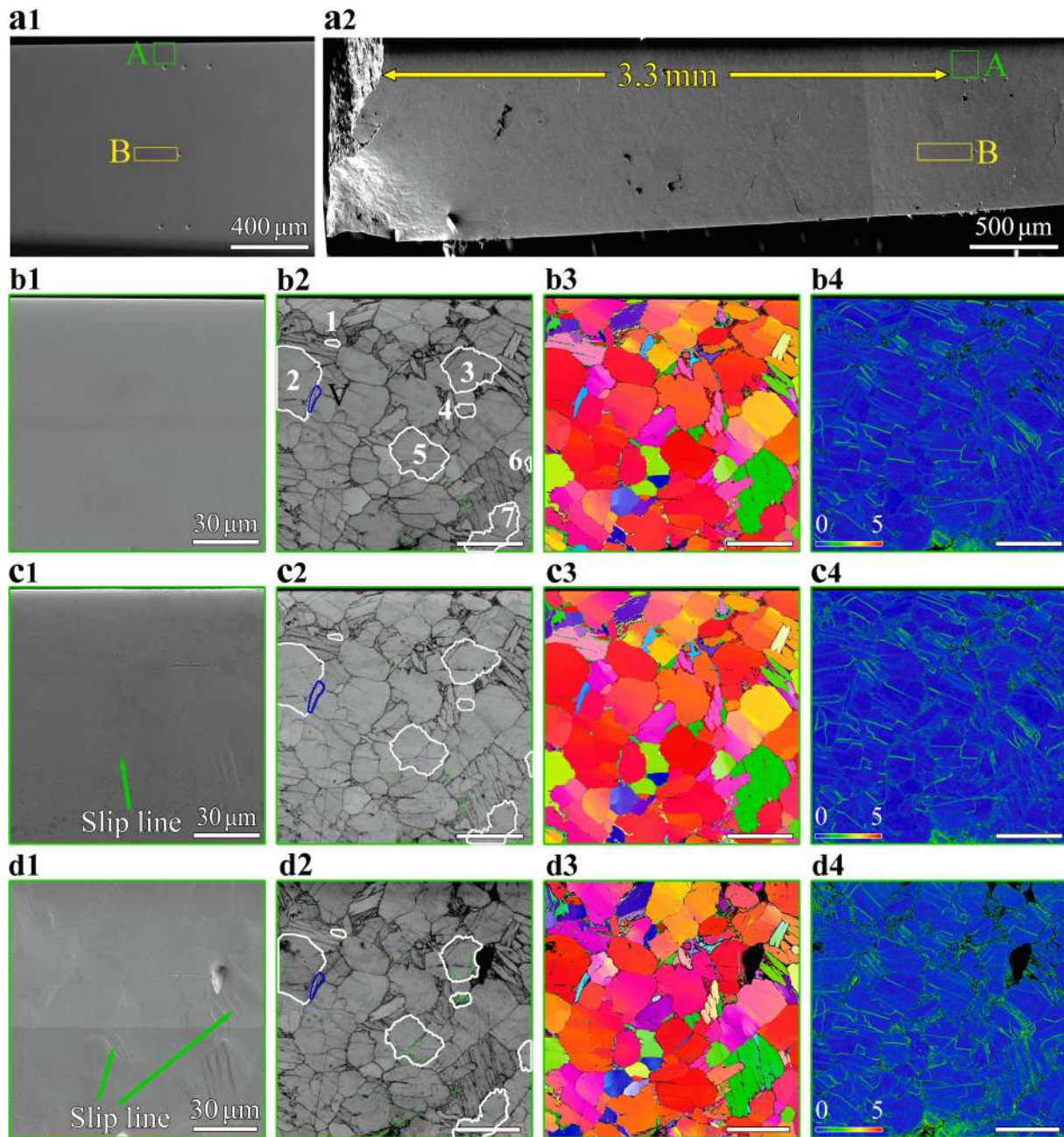


Fig. 9. Microstructure evolution during fatigue loadings ($\sigma_{\max} = 815$ MPa and $N_f = 10686$ cycles). a1 and a2: SEM observation of part of the initial specimen and the same specimen after 10,686 cycles, respectively. b1-b4, c1-c4 and d1-d4: Enlarged SEM image of the labelled region A and the associated BC map, IPF and KAM map of the initial specimen, the specimen after 883 and 10,686 cycles, respectively. The α grain enclosed by the white line in b2-d2 is with twinning occurrence, and the green lines denote the $\{10\bar{1}2\}$ twin boundary. The α grain enclosed by the blue line in b2-d2 is with bigger intracrystal deformation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

basal slip is bigger than the one for prismatic slip except for two grains for the α grains with the activation of basal slip and most of the SFs for basal slip are larger than 0.35. The SF for prismatic slip is bigger than the SF for basal slip except for one grain for the α grains with the activation of prismatic slip and the SFs for prismatic slip are often bigger than 0.35.

Similarly, the activation of multi slip systems of α grains in the same regions in Fig. 13a-13c are analyzed after the failure of the specimen. The results are shown in Fig. 13g-13i. The associated SFs for basal slip, prismatic slip and pyramidal slip in α grains are shown in Fig. 13j-13l, respectively. It is seen from Fig. 13g and 13j that there are three α grains activate the multi slip systems for the specimen under fatigue loadings. One is the basal slip and pyramidal slip, and the other two are basal slip and prismatic slip. Different from the single slip system in α grains, the

SFs of the prismatic slip are smaller for the α grains with multi slip systems. For the dwell fatigue with dwell time of 10 s, the multi slip of basal slip and prismatic slip dominates, and most of the SFs for prismatic slip are smaller than 0.35, as shown in Fig. 13h and 13k. For the dwell fatigue with dwell time of 120 s, the multi slip systems activate in two α grains. They are both prismatic slip and pyramidal slip. The single and multi slip systems activated at 883 loading cycles are also indicated for the fatigue specimen in Fig. 13d and 13g, which further shows the evidence that the slip system tends to activate at first in α grains with bigger SFs for basal slip or prismatic slip during fatigue loadings.

The results in Fig. 13 show that the basal slip and prismatic slip dominate in α grains for both the fatigue and dwell fatigue loadings and it tends to activate the slip system with bigger SFs for basal slip or

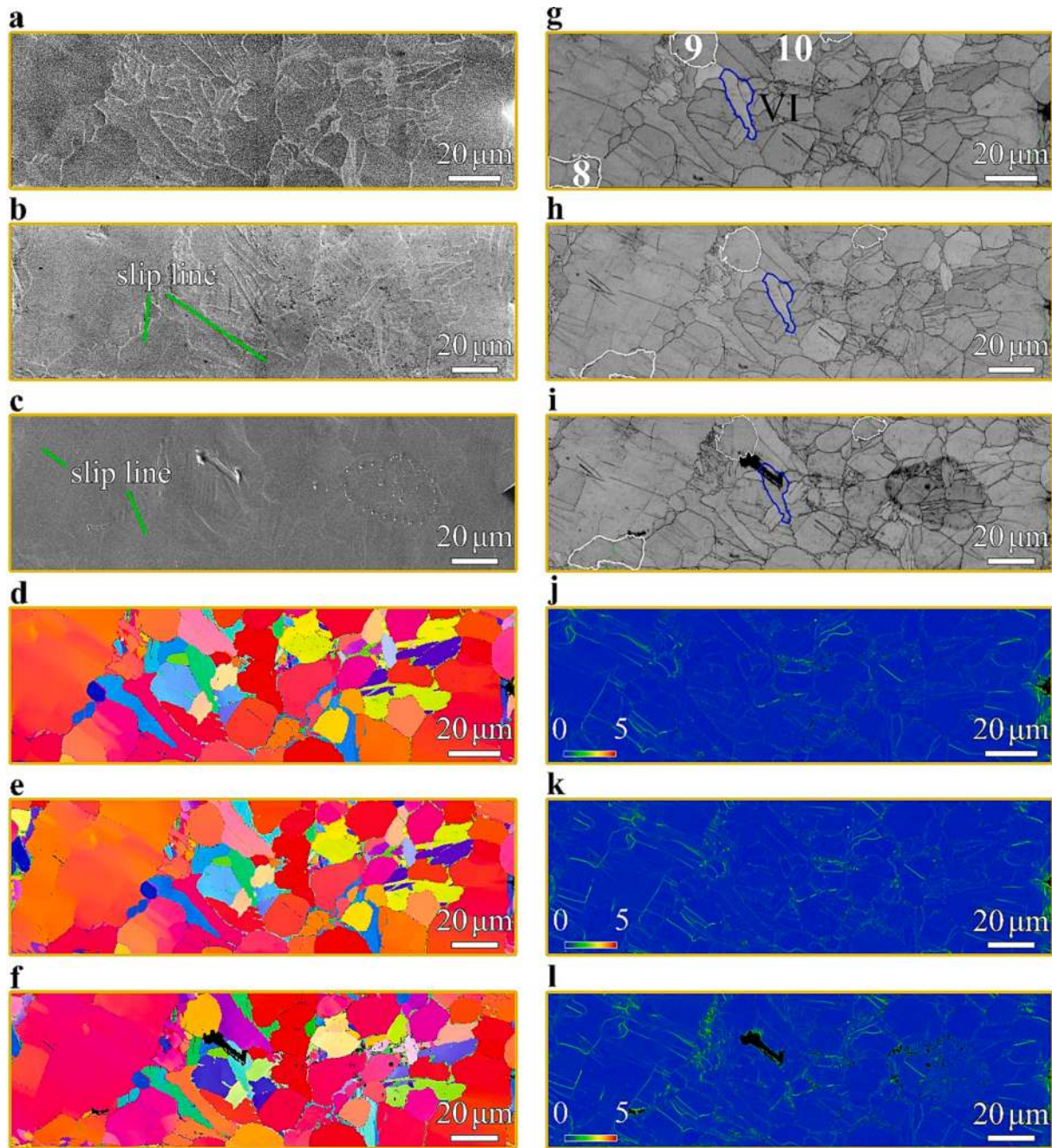


Fig. 10. SEM and EBSD results of the labelled region B in Fig. 9a during fatigue loadings ($\sigma_{\max} = 815$ MPa and $N_f = 10686$ cycles). a-c: Enlarged SEM image of the initial specimen, the specimen after 883 cycles and after 10,686 cycles, respectively. d-f, g-i and j-l: The associated IPF, BC map and KAM map in a-c, respectively. The α grain enclosed by the white line in g-i is with twinning occurrence, and the green lines denote the $\{10\bar{1}2\}$ twin boundary. The α grain enclosed by the blue line in g-i is with bigger intracrystal deformation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

prismatic slip. With the increase of loading cycles (i.e., the deformation) and further deformation compatibility, the slip systems with relative smaller SFs for basal slip or prismatic slip could occur due to microstructure inhomogeneities. In other words, the slip does not always occur on the basal plane or prismatic plane with large SFs in α grains, which depends on the deformation and the microstructure around the α phase, as indicated in Fig. 13d, 13e, 13j and 13k. Moreover, the occurrence of multi slip systems in α grains under fatigue loadings and dwell fatigue loadings with the maximum stress dwell of 120 s is much less than that under dwell fatigue loadings with the maximum stress dwell of 10 s. This indicates that a certain degree of maximum stress dwell favors the activation of slip system, but the occurrence of multi

slip does not increase with the increase of the dwell time of the maximum stress, i.e., it is not the greater the deformation the more favorable for the slip. This might be due to that parts of the crystal lattice distort with the increase of deformation and the sliding is hard to occur in grains with bigger lattice distortion or misorientation. It is supported by the observation of the dwell fatigue specimen under the maximum stress dwell of 120 s that the activation of slip is not increased from 74 to 199 loading cycles in Fig. 13f and 13l.

4.3. Twinning under fatigue and dwell fatigue loadings

The twins are observed in α grains during fatigue and dwell fatigue

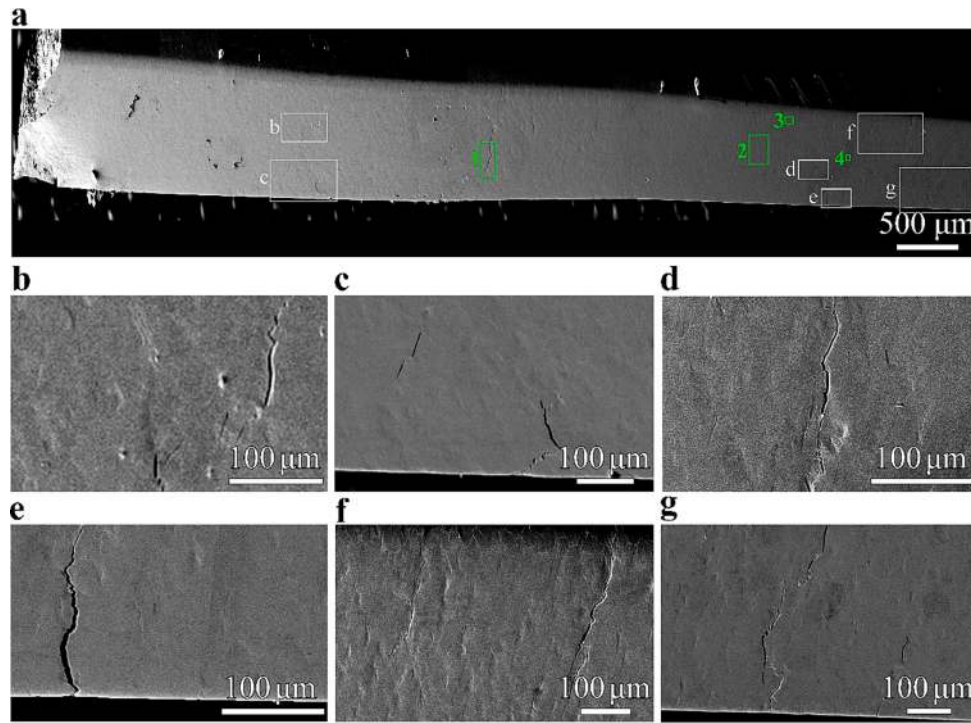


Fig. 11. SEM observation of the specimen surface after 10,686 fatigue cycles ($\sigma_{\max} = 815$ MPa and $N_f = 10686$ cycles). a: Part of the specimen surface. b-g: Close-ups of the regions b-g in a, respectively. The regions 1 to 4 in a denote where the microstructure and crack are further analyzed in Sec. 4.5.

loadings, as shown in the α grains enclosed by the white lines in Figs. 3, 4, 6, 7, 9 and 10. They are all extension twins. The crystallographic orientation of the corresponding parent α grains is analyzed and shown in Fig. 14. For the fatigue and dwell fatigue with dwell time of 10 s, the information of crystallographic orientation is calculated from the original α grains due to the EBSD observation of these labelled regions before fatigue and dwell fatigue tests. While for the dwell fatigue with dwell time of 120 s, it is calculated from α grains after 74 loading cycles. It is seen from Fig. 14 that the SFs for basal slip are either bigger or smaller for the parent α grains, but all the SFs for prismatic slip are not bigger than 0.2 and the angle of c-axis to the applied axial stress is smaller than 46° under both fatigue loading and dwell fatigue loadings. This indicates that the twinning occurs in α grains with SFs not bigger than 0.2 for prismatic slip and their c-axis to the applied axial stress smaller than 46° , which is irrespective of the maximum stress dwell and dwell time. The occurrence of twinning in α grains with SFs not bigger than 0.2 for prismatic slip is also irrespective of whether the α grain near or far away from the fracture surface (i.e., the degree of deformation). It is noted that just parts of the parent α grains with twinning occurrence are analyzed for crystallographic orientation in the labelled region A near the fracture surface in Fig. 6.

Further, the results of the fatigue specimen in Figs. 9 and 10 and those of the dwell fatigue specimen in Figs. 3 and 4 at the same loading cycles (i.e., 883 cycles for the fatigue and dwell fatigue with dwell time of 10 s) indicate that the maximum stress dwell could promote the twinning occurrence in α grains at the same loading cycles. While for almost the same accumulative dwell time (883 cycles under the dwell fatigue loading with dwell time of 10 s and 74 cycles under the dwell fatigue loading with dwell time of 120 s), the maximum stress dwell with dwell time of 120 s does not result in the twinning occurrence in more α grains in Fig. 7 than the maximum stress dwell with dwell time of 10 s in Figs. 3 and 4. This indicates that whether the twinning can occur depends on both the crystallographic orientation of the grains and the loading condition. The local high stress or strain is only the motivation for the occurrence of twinning. Relative to the fatigue loading, the accumulation of local strain or deformation is bigger under dwell fatigue

loadings, which can activate the occurrence of twinning in more α grains that are capable of twinning. This explains the reason that the maximum stress dwell with dwell time of 10 s activate the occurrence of twinning in more α grains in Figs. 3 and 4 compared with the fatigue loading in Figs. 9 and 10. It also explains the reason that the maximum stress dwell with dwell time of 120 s leads to the twinning occurrence in less α grains in Fig. 7 than the maximum stress dwell with dwell time of 10 s in Figs. 3 and 4 after the fracture of specimens. It is noted that the total area of the labelled regions A and B are both $2.5 \times 10^4 \mu\text{m}^2$ for the fatigue specimen in Figs. 9 and 10 and the dwell fatigue specimen with dwell time of 10 s in Figs. 3 and 4. The area of the labelled region B is $3.5 \times 10^4 \mu\text{m}^2$ for the dwell fatigue specimen with dwell time of 120 s in Fig. 7. The difference of the number of α grains in the compared regions are not considered for this analysis.

4.4. Grain refinement under fatigue and dwell fatigue loadings

The results for very high cycle fatigue of Ti alloys have shown that nanograins could form during fatigue loadings, and twinning is one main contributor for nanograin formation [30]. For the present Ti-6Al-4V ELI alloy, the refined grains are found in some α grains after a number of fatigue and dwell fatigue loading cycles, as shown in Figs. 3, 4, 6, 7, 9 and 10. The refined grains observed are mostly the twin variants. Especially, the evolution process from subgrains with twin relation to subgrains without twin relation is observed in α grains under dwell fatigue loadings with dwell time of 120 s, as shown in Fig. 15. The twinning occurs in the α grain at or before 74 dwell fatigue loading cycles. With the increase of the loading cycle, the twin variants grow and the dislocation accumulation or deformation further leads to the change of the misorientation in twin variants, and finally causes the formation of subgrains without twinning relation in the original α grain, as shown in Fig. 15d. This indicates that twinning is one main contributor for the refined grain formation during the low cycle dwell fatigue. The present result for the first time gives a direct observation of the formation of refined grains due to twinning and the further deformation under dwell fatigue loadings.

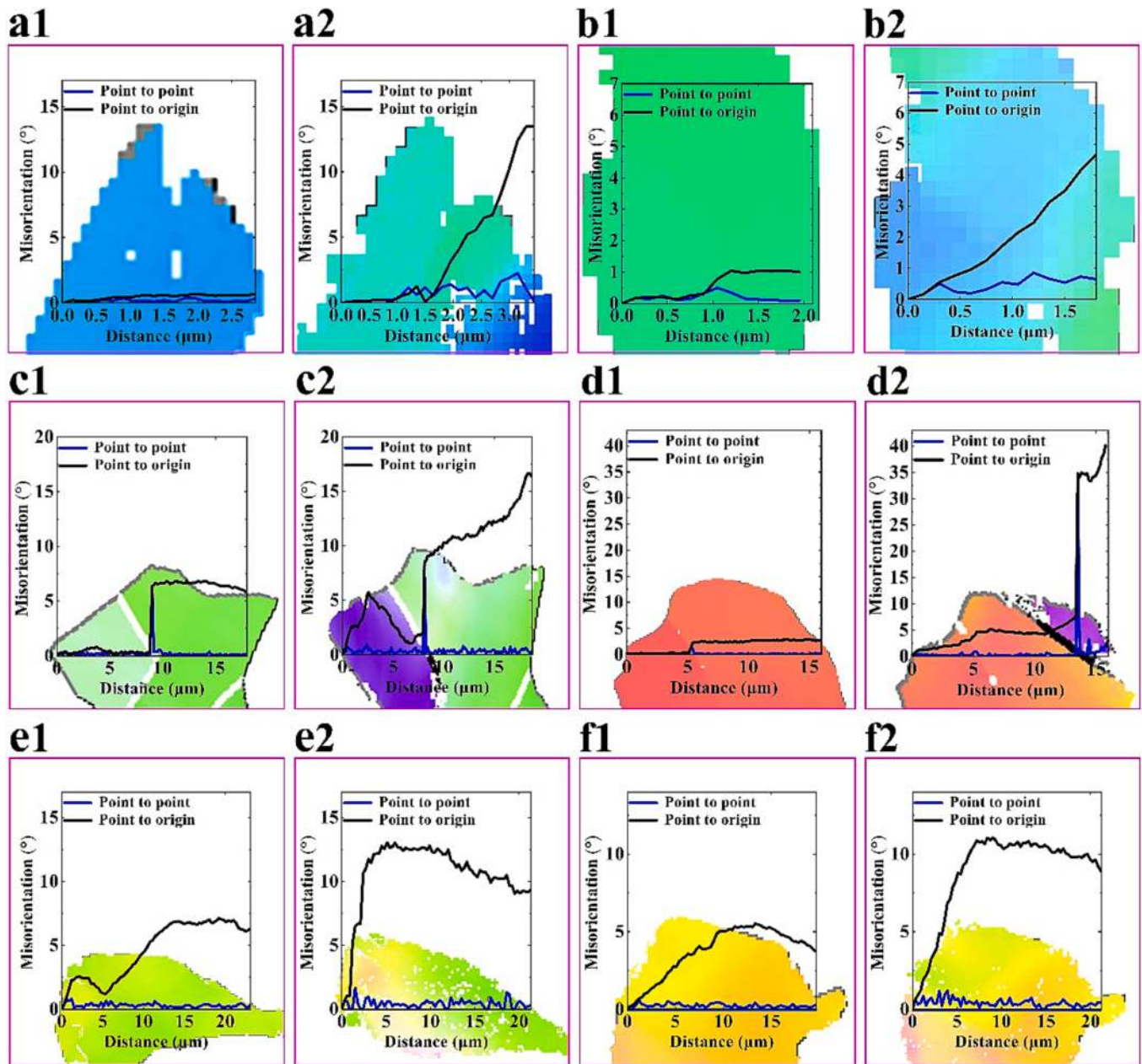


Fig. 12. Misorientation and subgrain formation in α grains during fatigue and dwell fatigue loadings. a1 and a2, b1 and b2: Part of grain V in Fig. 9b2 and part of grain VI in Fig. 10g during fatigue loadings, respectively; a1 and b1 are the initial grains without experiencing fatigue loadings, a2 and b2 are the grains after 10,686 cycles. c1 and c2, d1 and d2: Parts of grains I and II in Fig. 3d during dwell fatigue loadings with dwell time of 10 s, respectively; c1 and d1 are the initial grains without experiencing dwell fatigue loadings, c2 and d2 are the grains after 883 dwell fatigue cycles. e1 and e2, f1 and f2: Parts of grains III and IV in Fig. 6b2 during dwell fatigue loadings with dwell time of 120 s, respectively; e1 and f1 are the grains after 74 dwell fatigue cycles, e2 and f2 are the grains after 199 dwell fatigue cycles.

4.5. Crack formation under fatigue and dwell fatigue loadings

The results in Sec. 3.2 and 3.3 indicate that the maximum stress dwell has significant influence on the microcrack formation and growth during cyclic loadings. To further understand the formation of microcracks and the influence of dwell stress on the microcrack nucleation and growth, some microcracks far from the fracture surface are analyzed for the fatigue specimen and dwell fatigue specimen with dwell time of 10 s. The microcracks analyzed are shown in Figs. 16 and 17. EBSD results of the microcrack within one grain (Fig. 16a1-16a4, Fig. 17b1-17b4 and 17c1-17c4) or mainly along the boundaries of two grains (Fig. 16d1-16d4 and 16f1-16f4) indicate that the microcracks under fatigue loading tend to initiate from the prismatic slip of α grains with

bigger SFs for prismatic slip, while the microcracks under dwell fatigue loading tend to initiate from the basal slip of α grains with bigger SFs for basal slip or from the grain boundaries of α grains with small angle of the c -axis to the applied axial stress, as shown in Fig. 18a and 18b.

Further EBSD analysis indicates that there are twins along the crack path and at the crack tip for some cracks under fatigue loadings and dwell fatigue loadings with dwell time of 10 s. The SFs of basal slip and prismatic slip are analyzed for the parent α grains and twin variants just at the crack tip and at the crack path of the microcrack, and the results are shown in Fig. 18c and 18d. It is seen that the SF of prismatic slip for twin variants is generally bigger than that of the parent grain under both dwell fatigue and fatigue loadings (except one for dwell fatigue and three for fatigue). The SFs of prismatic slip for twin variants are

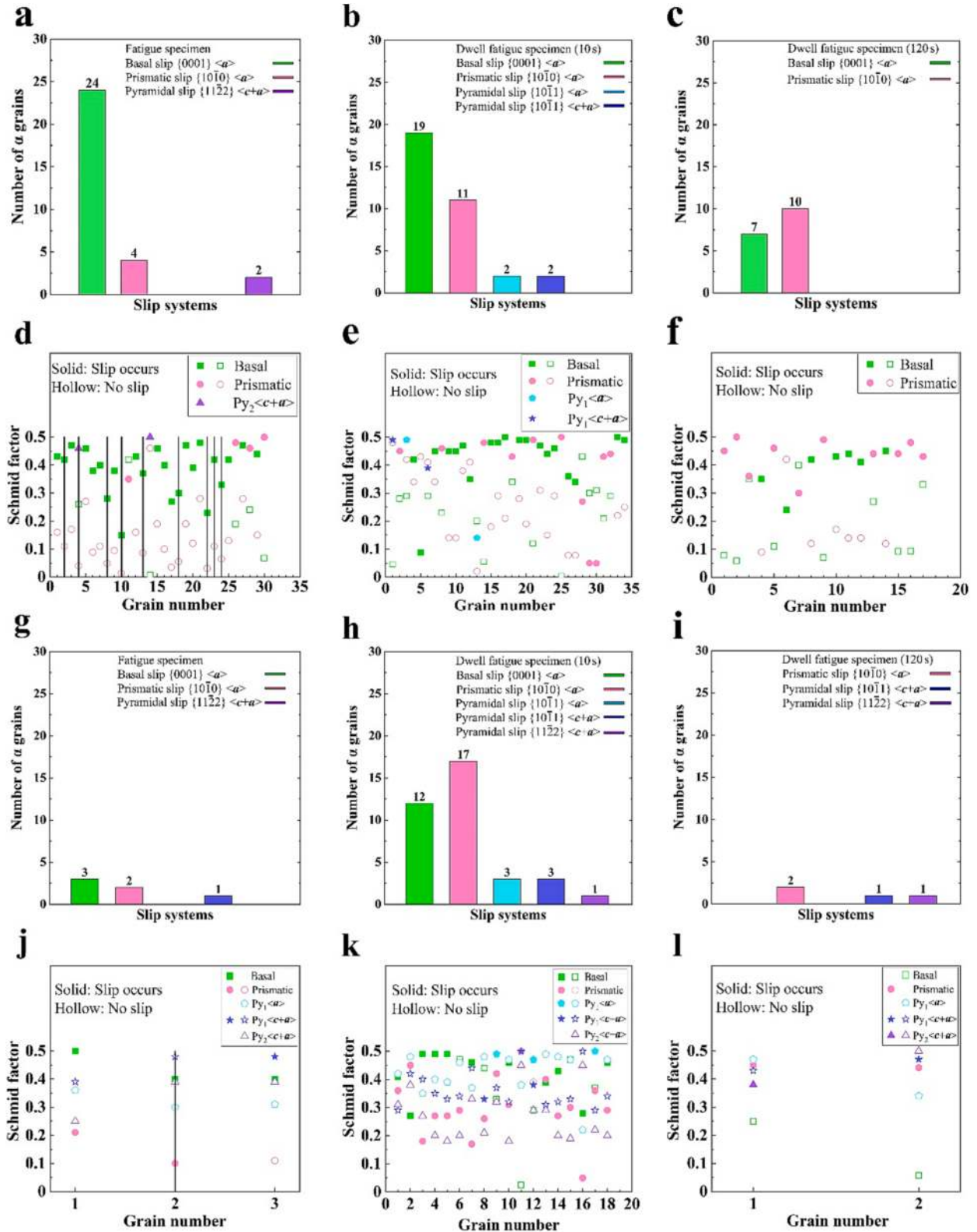


Fig. 13. Activated slip systems and SFs for basal slip, prismatic slip and pyramidal slip of α grains at specimen surface under fatigue and dwell fatigue loadings. a-c: Activation of single slip systems in labelled regions A and B in Fig. 9a1, labelled regions A and B in Fig. 3a and the labelled region B in Fig. 6a, respectively. d-f: SFs for basal slip, prismatic slip and activated pyramidal slip of α grains in a-c, respectively, in which the lines in d denote that these slip systems are not activated at 883 cycles for the fatigue specimen. g-i: Activation of multi slip systems in labelled regions A and B in Fig. 9a1, labelled regions A and B in Fig. 3a and the labelled region B in Fig. 6a, respectively. j-l: SFs for basal slip, prismatic slip, and activated pyramidal slip of α grains in g-i, respectively, in which the line in j denotes that the prismatic slip is not activated at 883 cycles for the fatigue specimen.

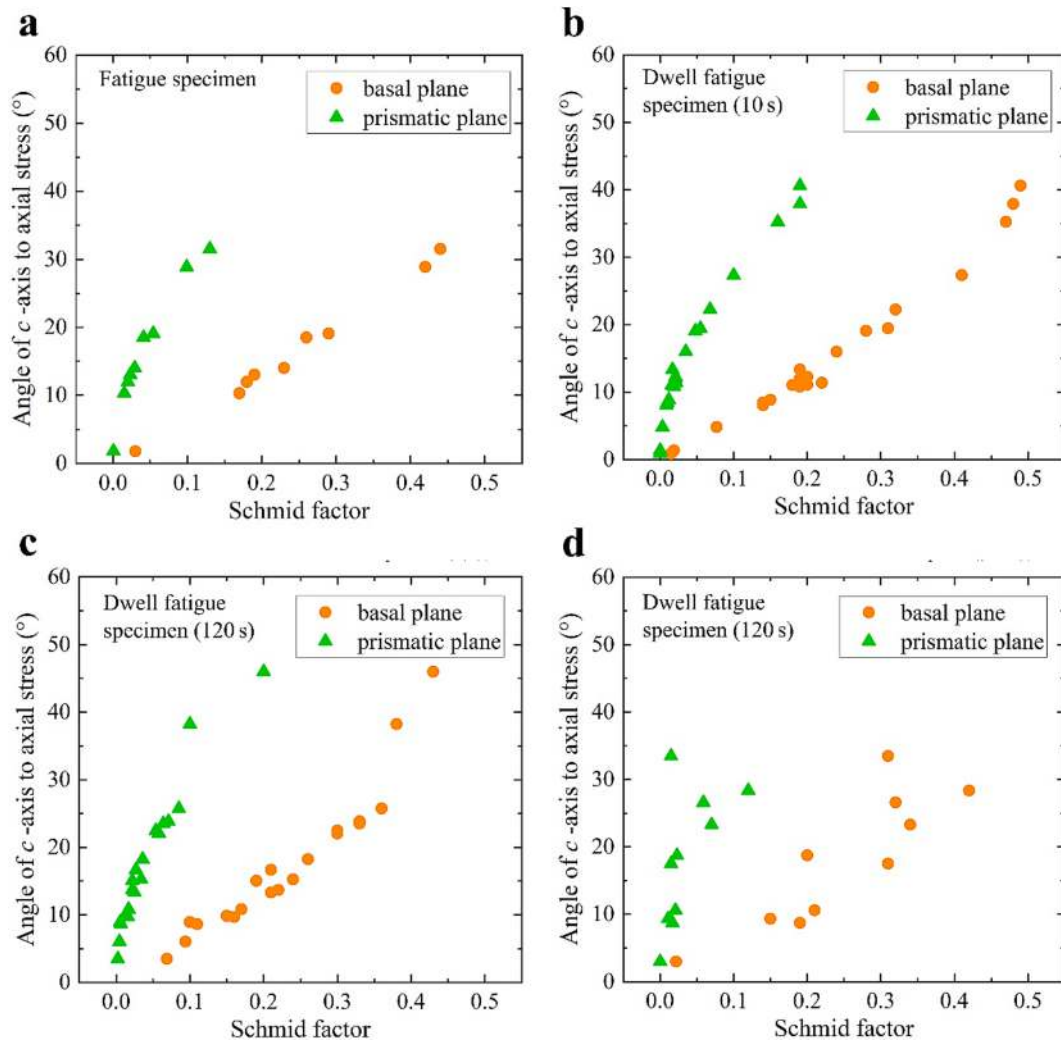


Fig. 14. SFs for basal slip, SFs for prismatic slip and the angle of c -axis to the applied axial stress of identifiable parent α grains during fatigue and dwell fatigue loadings. a: Parent α grains in labelled regions A and B in Fig. 9a1 far away from the fracture surface; b: Parent α grains in the labelled regions A and B in Fig. 3a far away from the fracture surface; c and d: Parent α grains in the labelled region B far away from the fracture surface and labelled region A near the fracture surface in Fig. 6a, respectively.

generally bigger than 0.4. For the SF of prismatic slip of the twin variant smaller than that of the parent grain under dwell fatigue loadings, its SF of basal slip is very large (bigger than 0.45). For the SF of prismatic slip for the three twin variants smaller than that of the associated parent grain under fatigue loadings, the SFs of basal slip of two twin variants are very large (bigger than 0.45). The SF of prismatic slip for twin variants near the crack surface or at crack tip is smaller than that of the parent grain is also observed for very high cycle fatigue of TC17 titanium alloy [27].

The SFs for basal slip and prismatic slip are also analyzed for identifiable twin variants and parent α grains in the quasi in-situ regions under dwell fatigue loadings and fatigue loadings. The results under dwell fatigue loadings with dwell time of 10 s are shown in Fig. 18e, and the results under the fatigue loading are shown in Fig. 18f. The results under dwell fatigue loadings with dwell time of 120 s are shown in Fig. 18g and 18h. For the α grain with the occurrence of more than one twinning, the SFs for basal slip and prismatic slip of one of these twin variants are chosen due to the small difference of the SFs for basal slip and prismatic slip between twin variants. It is seen that the SFs of prismatic slip for twin variants are all bigger than 0.35 and bigger than the SFs of prismatic slip of their parent grains under both dwell fatigue and fatigue loadings, which is irrespective of the observed region far away or near the fracture surface. This indicates that the occurrence of

twinning favors deformation due to the increase of SFs for prismatic slip. Combined the results in Fig. 18c and 18d, the relative smaller or very small SF of prismatic slip of the twin variant at crack tip or crack path of the microcrack should be attribute to the large deformation at crack tip or during the crack growth.

A comparison of the results in Fig. 16 under dwell fatigue loadings and those in Fig. 17 under fatigue loadings indicates that the twinning often occurs in α grains at microcrack tip under dwell fatigue loadings, namely that the maximum stress dwell increases the plastic deformation and tends to cause the occurrence of twinning at microcrack tip. Twinning releases the stress intensification at crack tip and favors the strain compatibility [31,32]. In other words, the plastic deformation due to the maximum stress dwell restrains the crack growth. This explains the much smaller cracks in Fig. 5 under dwell fatigue loadings with dwell time of 10 s than those in Fig. 11 under fatigue loadings. For the dwell fatigue loading with dwell time of 120 s without crack observed on the specimen surface far from the fracture surface after the specimen fails, it is thought that the overall deformation of the specimen is large and the crack induced by sliding or cleavage (i.e., brittle crack) can hardly occur in grains with distorted lattice due to deformation. In this case, the local high deformation or strain accumulation tends to cause the formation of cavity [17], as shown in Fig. 8f. The cavity increases or some cavities coalesce, and finally lead to the failure of the specimen.

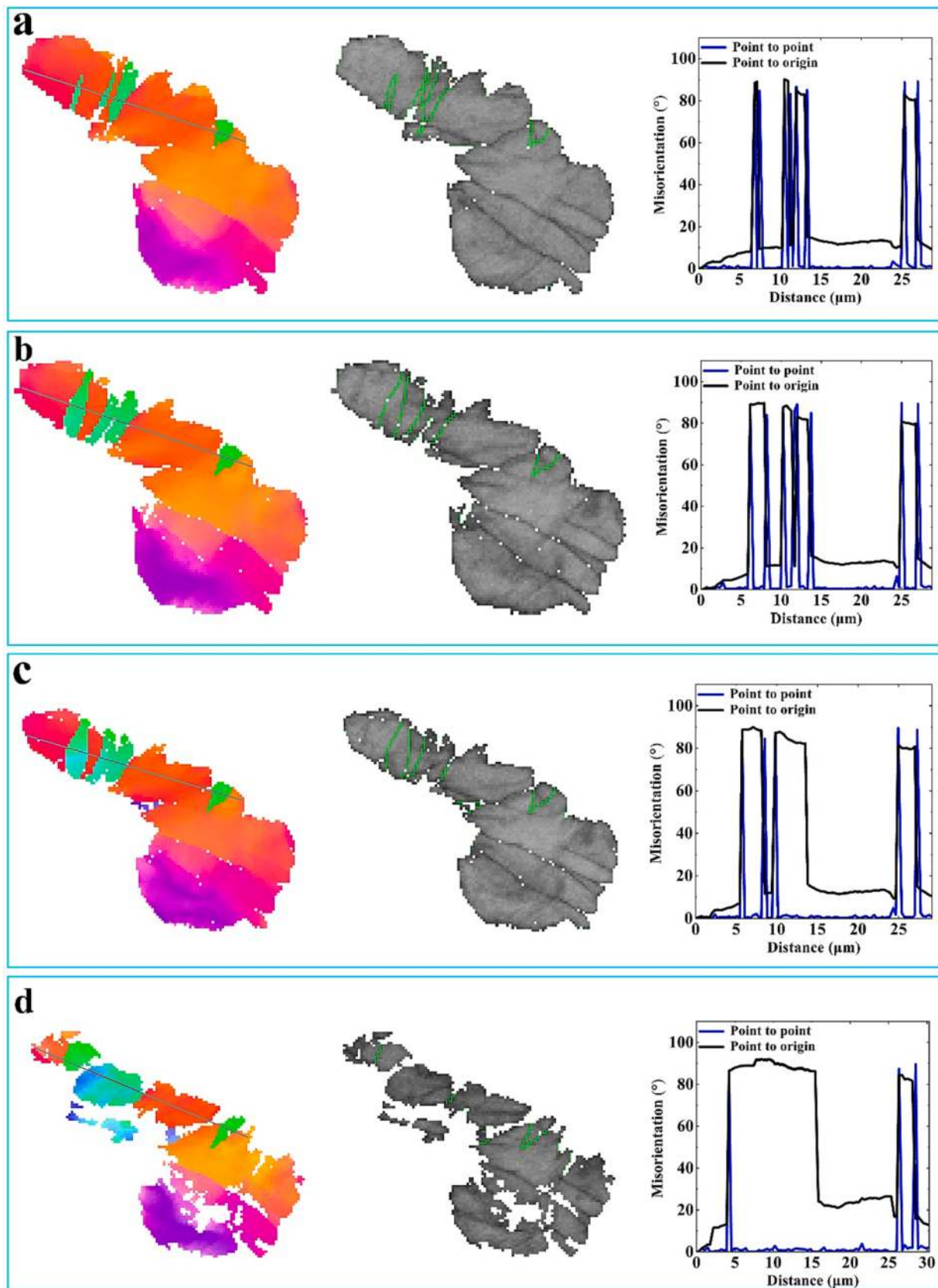


Fig. 15. Evolution process of the refined grain in α grain 1 in Fig. 6 during dwell fatigue loadings with dwell time of 120 s. a-d: IPF (left), BC map (middle) and misorientation (right) on the line path in IPF after 74, 126, 178 and 199 loading cycles, respectively. The green lines in BC maps denote the $\{10\bar{1}2\}$ twin boundary. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

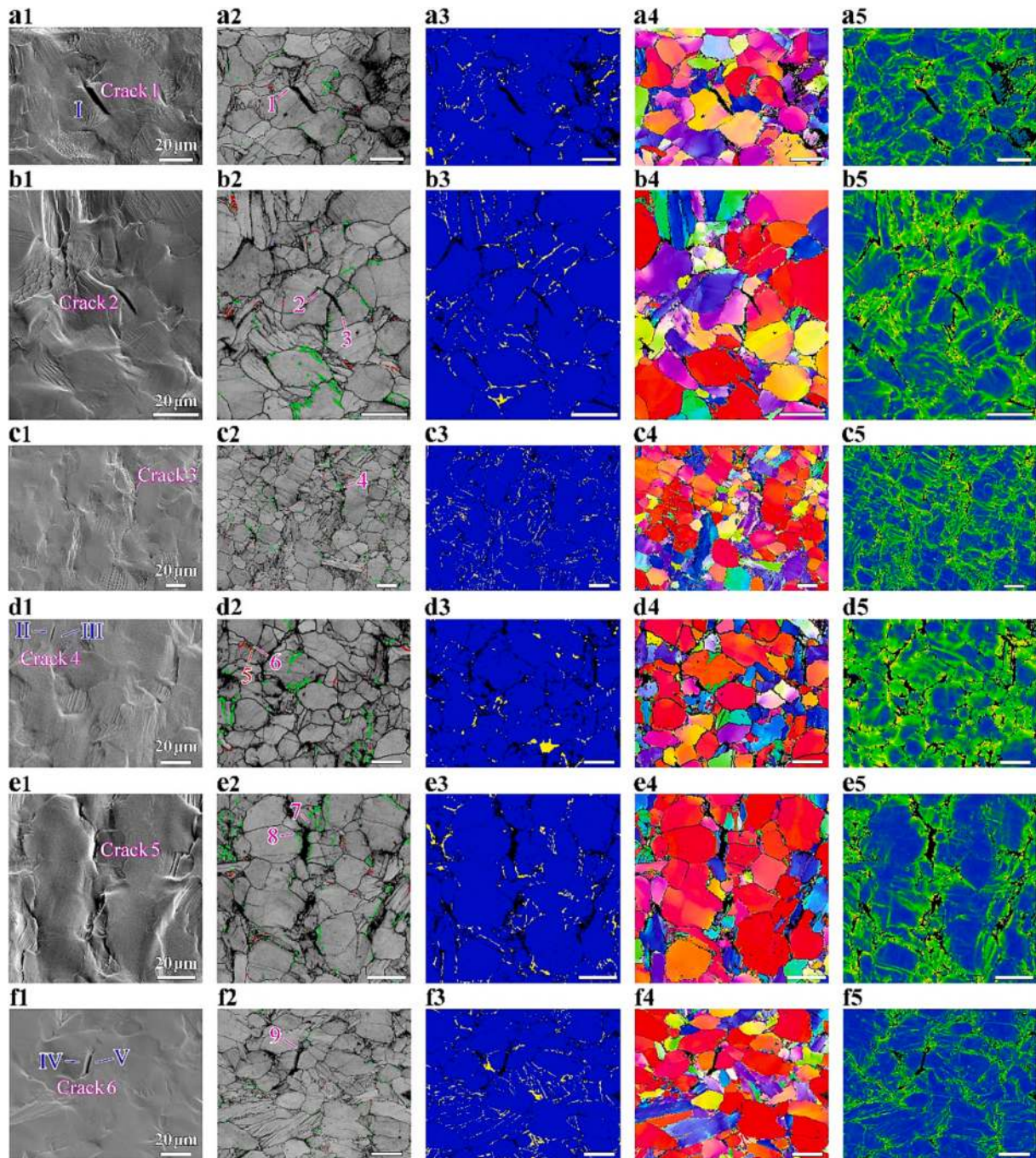


Fig. 16. Microcracks at the specimen surface after 883 dwell fatigue loading cycles ($\sigma_{\max} = 815$ MPa, $d_{t,\max} = 10$ s and $N_f = 883$ cycles). a1-a5, b1-b5, c1-c5, d1-d5 and e1-e5: Enlarged SEM image of regions 1 to 5 and the associated BC map, phase map, IPF and KAM map in Fig. 5a, respectively. f1-f5: SEM image of the local region containing the microcrack in Fig. 4b and the associated BC map, phase map, IPF and KAM map, respectively. The green and red lines in a2-f2 denote the $\{10\bar{1}2\}$ twin boundary and $\{10\bar{1}1\}$ twin boundary, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.6. Failure mechanism under dwell fatigue loadings

The results in Sec. 4.5 indicate that the fatigue loading is an important factor for the brittle crack nucleus though the dwell of the maximum stress increases the plastic strain accumulation in α grains and at phase boundaries, as indicated in the KAM values in Figs. 3, 4, 7, 9 and 10. A certain degree of maximum stress dwell increases the plastic strain accumulation and favors the brittle crack nucleation [13,15]. However, if the dwell stress is high or the dwell time is relative bigger, the plastic strain induced by the dwell stress could restrain the microcrack growth

and result in the ductile failure mode shown in Fig. 2a-2f [8]. Therefore, the dwell fatigue failure of titanium alloys could exhibit the following three mechanisms due to the difference of dwell stress and dwell time:

i) *Fatigue failure:* The crack initiation and growth dominate the failure. In this case, the deformation or strain due to the dwell stress promotes the brittle crack initiation and growth, or the dwell stress is very small and the influence of dwell stress is very small or negligible. The fracture surface presents a large region of crack initiation and growth feature (Fig. 2g-2i).

ii) *Ductile failure:* The plastic deformation dominates the failure. In

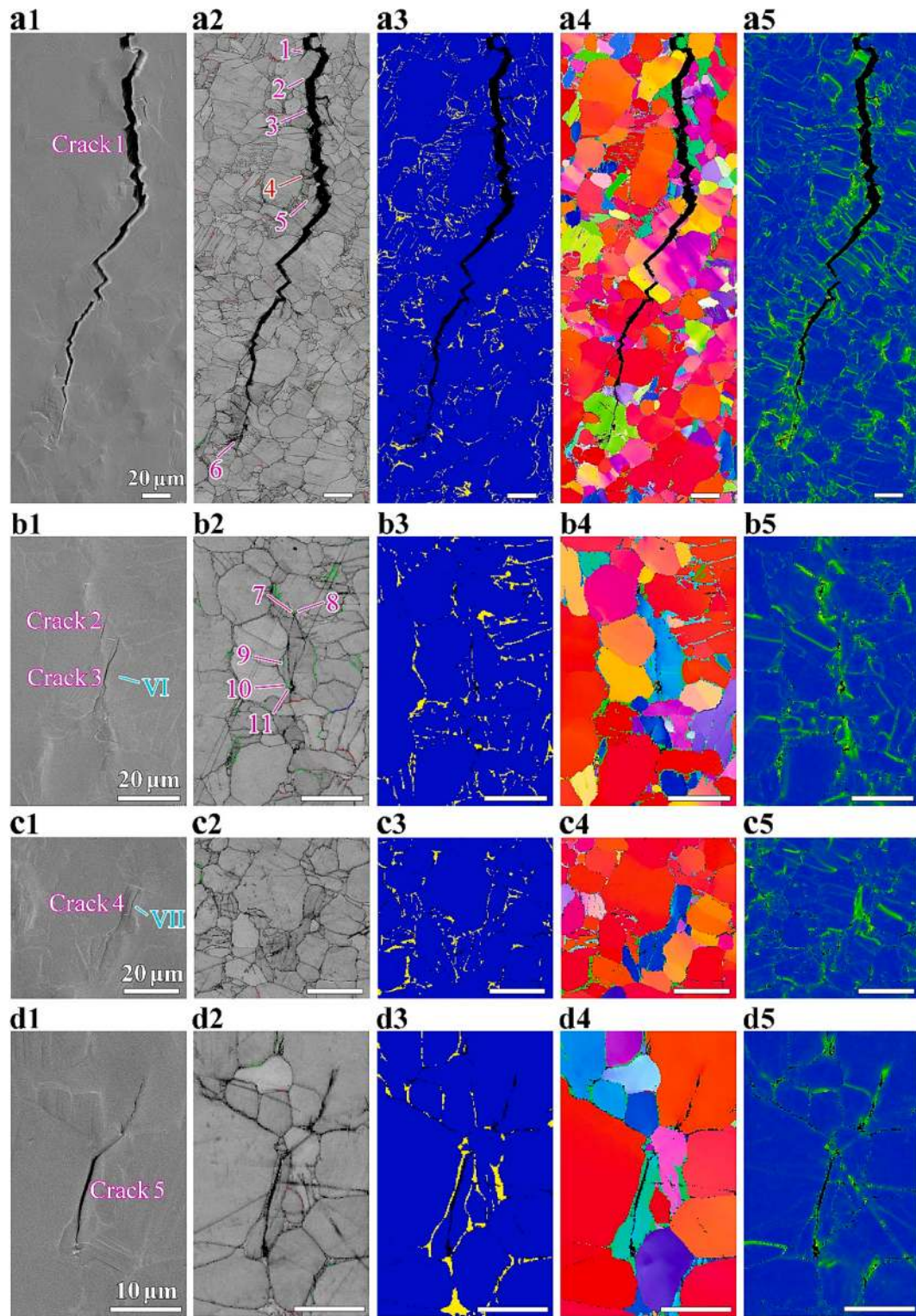


Fig. 17. Microcracks at the specimen surface after 10,686 fatigue loading cycles ($\sigma_{\max} = 815$ MPa and $N_f = 10686$ cycles). a1-a5, b1-b5, c1-c5 and d1-d5: Enlarged SEM image of regions 1 to 4 and the associated BC map, phase map, IPF and KAM map in Fig. 11a, respectively. The green and red lines in a2-d2 denote the $\{10\bar{1}2\}$ twin boundary and $\{10\bar{1}1\}$ twin boundary, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

this case, the dwell stress is high or the dwell time is long, no brittle crack initiation or very small regions of brittle crack initiation and growth form due to the bigger plastic deformation. The fracture surface presents no or very small regions of brittle crack initiation and growth and dimple like features (Fig. 2a-2f). It is noted that the ductile failure

can also occur under fatigue loadings due to the high stress or stress ratio [33,34], in which the part of the higher stress in the fatigue loading causes the plastic deformation accumulation and results in the ductile failure.

iii) *Mixed failure:* The brittle crack initiation and growth and the

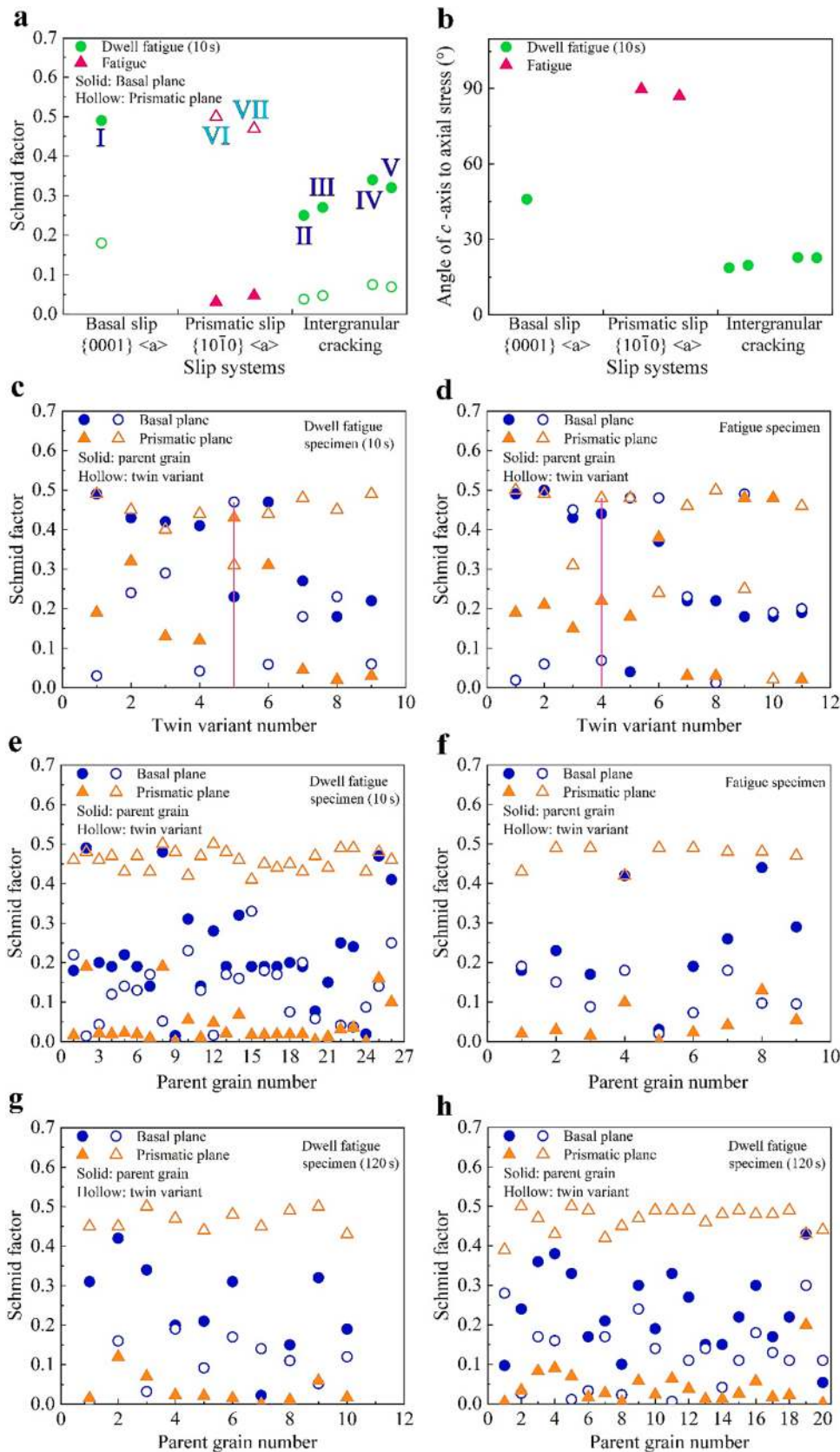


Fig. 18. SFs and angle of the c-axis to the applied axial stress of α grains for the fatigue specimen and dwell fatigue specimen. a: SFs for basal slip and prismatic slip of numbered α grains in Fig. 16a1, 16d1, 16f1, 17b1 and 17c1 for crack initiation. b: Angle of the c-axis to the applied axial stress of α grains for crack initiation in a. c and d: SFs for basal slip and prismatic slip of the numbered twin variants and the associated parent α grains in Fig. 16a2-16f2 and in Fig. 17a2 and 17b2, respectively, in which the pink solid lines represent compression twins and the others are extension twins. e and f: SFs for basal slip and prismatic slip of identifiable twin variants and parent α grains in Fig. 3h and 4d and in Fig. 9d2 and 10i, respectively, in which the horizontal ordinates in e and f denote the parent α grain number in Fig. 3d and 4c and in Fig. 9b2 and 10g, respectively. g and h: SFs for basal slip and prismatic slip of identifiable twin variants and parent α grains in Fig. 6e2 and 7d2, respectively, in which the horizontal ordinates denote the parent α grain number in Fig. 6b2 and 7a2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

plastic deformation both play an important role in the final failure. In this case, the specimen exhibits brittle crack initiation and growth in some local regions at the fracture surface, but the fracture is due to both the brittle crack initiation and growth and the plastic deformation induced cracking [8,35].

The above results indicate that the dwell fatigue behavior should be analyzed from the view of influence of dwell loading on the fatigue property for the fatigue failure. While for the ductile failure, one should analyze the dwell fatigue behavior from the view of influence of fatigue loading on the creep property. The above results also indicate that the fatigue life cannot be effectively predicted through the traditional crack initiation and growth rate for ductile or mixed failure due to that it is not the failure dominated by the brittle crack initiation and growth.

5. Conclusions

In this paper, a method of quasi in-situ EBSD observation is developed by the combination of the conventional testing machine (e.g., MTS machine) and EBSD observation. Then, it is used to investigate the microstructure and damage evolution of the Ti-6Al-4V ELI alloy used for pressure hulls of submersibles under fatigue and dwell fatigue loadings. The main results are as follows:

- (1) The slip system tends to be activated in α grains with bigger SFs for basal slip or prismatic slip, and the basal slip or prismatic slip dominates for both fatigue and dwell fatigue of the Ti-6Al-4V ELI alloy. A certain degree of maximum stress dwell favors the activation of slip system, but it is not the greater the deformation, the more favorable the slip system activation. The increment of misorientation and the subgrain formation are captured in α grains under both fatigue and dwell fatigue loadings, which provides with the evidence of grain refinement due to dislocation sliding and dislocation interaction during cyclic loadings.
- (2) Twinning occurs in α grains with SFs not bigger than 0.2 for prismatic slip and their c -axis to the applied axial stress smaller than 46° , which is irrespective of the maximum stress dwell and dwell time. Whether the twinning occurs or not in α grains depends on both the crystallographic orientation of grains and loading condition. A certain degree of dwell stress promotes the occurrence of twinning in more α grains that are capable of twinning. The process of subgrain formation in α grains due to twinning is observed, which indicates that twinning is one main contributor for grain refinement under cyclic loadings.
- (3) Microcracks tend to initiate from prismatic plane of α grains with bigger SFs for prismatic slip under fatigue loadings, while microcracks tend to initiate from basal plane of α grains with bigger SFs for basal slip or boundaries of α grains with small angle between the c -axis and the applied axial stress under dwell fatigue loadings. The maximum stress dwell has significant influence on the brittle crack initiation and growth. A certain degree of maximum stress dwell increases the plastic strain accumulation and favors the brittle crack formation. However, if the dwell stress is high or the dwell time is long, the plastic strain induced by the dwell stress could restrain the brittle microcrack growth and result in the ductile failure mode.

The paper presents an insight into the influence of maximum stress dwell and dwell time on the crack nucleus and damage evolution of Ti alloys in microstructural scale. It explains the different dwell fatigue mechanism due to the difference of dwell stress level and dwell time.

CRedit authorship contribution statement

Chengqi Sun: Methodology, Visualization, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Supervision. **Jian Sun:** Visualization, Formal analysis, Investigation, Writing –

review & editing. **Weiqlan Chi:** Methodology, Visualization, Formal analysis, Investigation, Writing – review & editing. **Jiaxuan Wang:** Investigation, Writing – review & editing. **Wenjing Wang:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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